Testing My Custom PRNG with TestU01 BigCrush

I recently put a custom pseudorandom number generator (PRNG) design through the **TestU01 BigCrush** battery—the industry standard for testing statistical randomness.

Here is the core logic prototype implemented in MATLAB/Octave ( ):

```
function [x,y] = prng21(x,y)
   n = 2^12; % Vector length (N = 4096)
   if nargin == 0
      x = rand(n,1); % Initial state vectors
      y = rand(n,1); % or true random numbers
   end
   % Coupled addition modulo 1
   x = mod(y + x, 1);
   y = mod(y + x, 1);
   % Cross-permutation using sort indices
   [~, idx_x] = sort(x);
   [~, idx_y] = sort(y);
   c = x;
   x = y(idx_x);
   y = c(idx_y);
end
```

To run the rigorous statistical tests efficiently, I ported this architecture to **C using 64-bit fixed-point integer arithmetic**.

The results were outstanding: the generator **passed the BigCrush battery (~160 tests)**!

#RandomNumberGenerator #PRNG #AlgorithmDesign #SoftwareEngineering #Clang #MATLAB #DataScience #TestU01 #ComputerScience #Coding

TestU01 BigCrush Evaluation Summary
Generator Configuration
Name / Variant: prng21 / prng30 (Vector size)
Framework: TestU01 1.2.3 (BigCrush Battery)
Total Statistics Tested: 160 per run
CPU Time per Run: 6 hours 55 minutes
Test Results Overview
Run
Key Findings / Deviations
Status
Run 1
159/160 passed normally. Test #13 (smarsa_BirthdaySpacings,) flagged a marginal deviation with . No catastrophic failures ( ).
PASSED (1 marginal suspect)
Run 2

BirthdaySpacings passed completely. Test #108 (swalk_RandomWalk1) flagged a borderline notice with (just below the boundary).
PASSED (1 borderline notice)
Detailed Evaluation
Pass Rate & Statistical Quality ( ):
Across both independent runs, the generator passed nearly of all 160 demanding statistical tests in TestU01's heaviest battery (BigCrush). It showed no catastrophic structural failures, which are typically indicated by extreme -values such as eps ( ) or eps1 ( ).

Expected Noise Profile:
In a test suite evaluating 160 distinct statistics under strict boundary rules ( ), purely ideal random sequences naturally produce boundary warnings due to standard probability distribution variance.

Conclusion

Verdict: BigCrush PASSED
The generator demonstrates exceptional statistical randomness, high dimensional independence, and robust period dynamics. It meets the highest criteria set by state-of-the-art PRNG statistical test batteries.